Thursday Reading Assessment: ADC and DAC (Chapter 12 + Lab)

Prof. Jordan C. Hanson

April 4, 2024

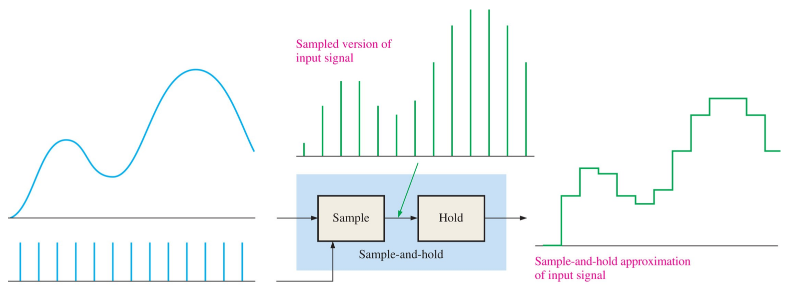


Figure 1: Simple picture of the sample-and-hold step in analog-to-digital conversion.

1 Analog-to-Digital Conversion

1. **Analog-to-digital conversion** (ADC) is a 2-step process by which analog signals are converted to digital data in digital circuits. The first step is depicted in Fig. 1 (top left, top right). In a **sample-and-hold** operation, the continuous signal is mixed with *sample pulses* emitted at rate equal to the sampling frequency. (Fig. 1 (top left)). The mixed sample pulses then cause a latch to *hold* the voltage value for the duration of the sample, $\Delta t = f_s^{-1}$ (Fig. 1 (top right)). (a) If Δt is 1 μs, what is the sampling frequency? (b) What is the Nyquist frequency?

2. In the next stage of ADC, the sample-and-hold output is *digitized*. Assume the space between maximum voltage, V_{CC}, and 0V, is quantized with N bits. (a) If $V_{\mathrm{CC}} = 5.0$ V, and $N = 3$ bits, what is the voltage resolution? (b) If $V_{\mathrm{CC}} = 12.0$ V, and $N = 12$ bits, what is the voltage resolution? (c) If we need at least 10 mV precision for $V_{\mathrm{CC}} = 5.0$ V, what is the minimum number of ADC bits?

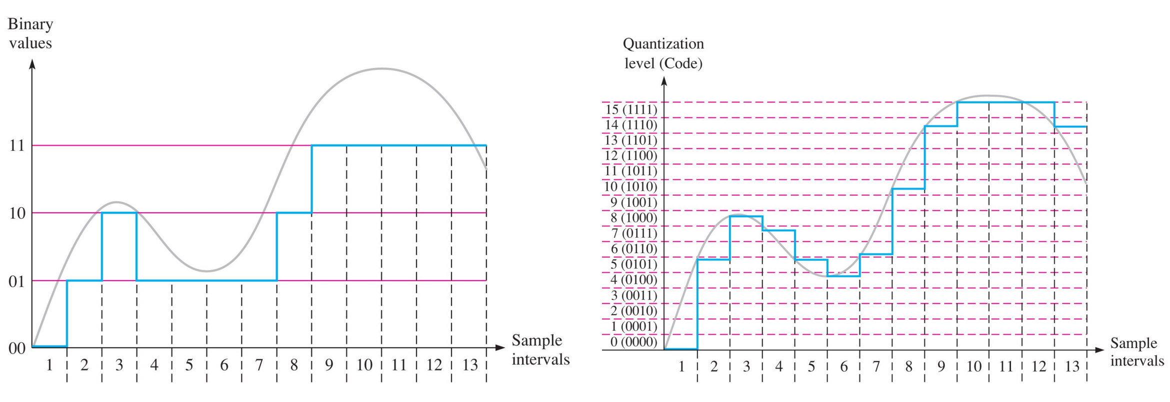
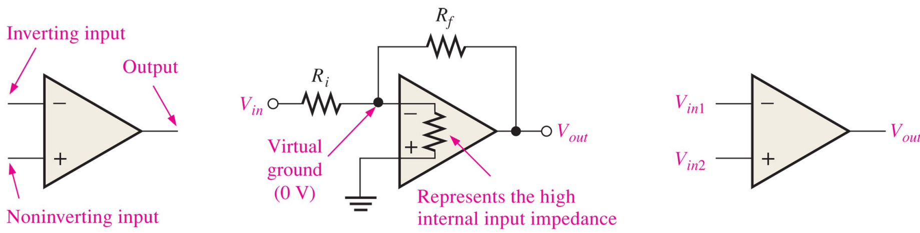


Figure 2: (Left) 2-bit ADC precision. (Right) 4-bit ADC precision.



(a) Op-amp symbol

(b) Op-amp as an inverting amplifier with gain of R_f/R_i

(c) Op-amp as a comparator

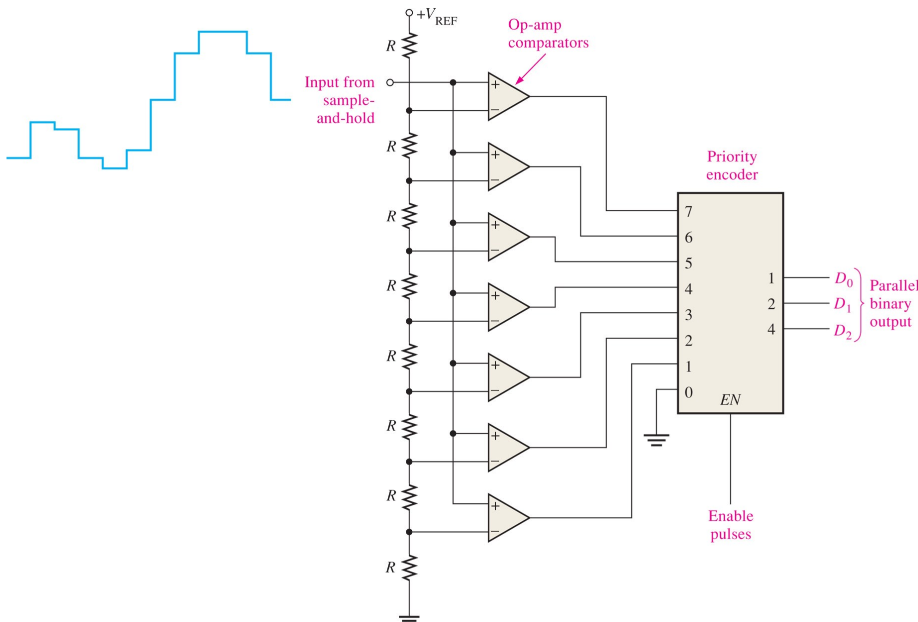


Figure 3: (Top) Op-amp characteristics. (Bottom) 3-bit flash ADC schematic.